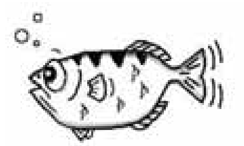
web based sophisticated bug tracker initially used in the development of Mozilla; GNATS – a Bug Tracking system initially used by GNU; Sourceforge and its forks provide separate PHP-based bug trackers and dedicated debugging forums for all their projects.



The GNU Debugger is a popular open source debugging tool

The conventional testing process used in proprietary software development falls short of the requirements of open source development, where integration is almost continuous and changes incremental and evolutionary.

Tinderbox allows any error in the newly written code to be detected live at the time of integration, through a continuous build process. It also detects the user responsible for the erroneous update and makes it his responsibility to rectify it, overall contributing to a highly reliable debugging process. Some other features desired in debuggers for open source development include remote debugging. This feature, found in the GNU Debugger used by the GNU project allows a developer to run the debugger over the internet.

The challengers have given rise to new debugging tools such as Tinderbox

Some processes that are common in the open source development community include partial rewrites and complete rewrites. Examples include the Apache web server, which was completely rewritten between version 1.3 and 2.0.

Testing of an open source project is intrinsically easier than that of proprietary software. With the users involved from a very early stage of development, bugs are detected and corrected early during development phase, often leading to extremely stable and bug free code. In addition, automated testing routines used in proprietary programs are also used in open source projects. The fact that most open source projects through a majority of their development phase use command line interfaces or application programming interfaces also makes automated testing easier.

While the various advantages of open source development are as mentioned above, success of a project largely depends